Disclaimer/credit. These questions have largely been ~~stolen~~ borrowed from natural sciences maths and vector calculus courses.

1) Let $\boldsymbol{a}$ and $\boldsymbol{b}$ be fixed vectors. Show that $\nabla(\boldsymbol{a}\cdot\boldsymbol{x})=\boldsymbol{a}$. Evaluate the divergence and the curl of each of the following vector functions of $\boldsymbol{x}$:

$$\mathbf{x}\,, \qquad \mathbf{a}(\mathbf{x}\cdot\mathbf{b})\,.$$

2) (i) Let $\phi(\mathbf{x})$ be a scalar field and $\mathbf{v}(\mathbf{x})$ a vector field. Show, using suffix notation, that

$$\nabla\cdot(\phi\mathbf{v})=\nabla\phi\cdot\mathbf{v}+\phi\,\nabla\cdot\mathbf{v}, \qquad \nabla\times(\phi\mathbf{v})=\nabla\phi\times\mathbf{v}+\phi\,\nabla\times\mathbf{v}.$$

(ii) Evaluate the divergence and curl of each of the following:

$$(\mathbf{a}\cdot\mathbf{x})\mathbf{b}, \qquad \mathbf{a}\times\mathbf{x}, \qquad r\mathbf{x}, \qquad \frac{\mathbf{x}-\mathbf{a}}{|\mathbf{x}-\mathbf{a}|^3},$$

where $r=|\mathbf{x}|$ and $\mathbf{a}$, $\mathbf{b}$ are constant vectors.

(iii) The vector fields $\mathbf{F}$ and $\mathbf{G}$ are everywhere parallel, non-zero and solenoidal. Show that $\mathbf{F}\cdot\nabla(F/G)=0$, where $F=|\mathbf{F}|$ and $G=|\mathbf{G}|$.

3) Use suffix notation to show that for vector fields $\mathbf{u}(\mathbf{x})$ and $\mathbf{v}(\mathbf{x})$,

$$\nabla\times(\mathbf{u}\times\mathbf{v})=(\nabla\cdot\mathbf{v})\mathbf{u}-(\nabla\cdot\mathbf{u})\mathbf{v}+(\mathbf{v}\cdot\nabla)\mathbf{u}-(\mathbf{u}\cdot\nabla)\mathbf{v}.$$

Show also that $(\mathbf{u}\cdot\nabla)\mathbf{u}=\nabla\big(\frac{1}{2}|\mathbf{u}|^2\big)-\mathbf{u}\times(\nabla\times\mathbf{u})$.

4) Trickier, note that solenoidal means $\nabla\cdot\mathbf{F}=0$.

Suppose that $\mathbf{F}\colon\mathbb{R}^3\to\mathbb{R}^3$ is a solenoidal vector field. Show that $\mathbf{F}=\nabla\times\mathbf{A}$ where

$$\mathbf{A}(\mathbf{x})=\int_0^1\mathbf{F}(t\mathbf{x})\times(t\mathbf{x})\,\mathrm{d}t.$$

5) For $r=|\mathbf{x}|$, verify, in index notation, that

$$\nabla r^n=nr^{n-2}\mathbf{x}.$$

6) For electric and magnetic fields $\mathbf{E}(\mathbf{x},t)$ and $\mathbf{B}(\mathbf{x},t)$, define the quantities $U=\frac{1}{2}\big(\epsilon_0|\mathbf{E}|^2+\mu_0^{-1}|\mathbf{B}|^2\big)$ and $\mathbf{P}=\mu_0^{-1}\mathbf{E}\times\mathbf{B}$. Use Maxwell's equations with $\mathbf{J}=\mathbf{0}$ to establish the conservation law $\partial U/\partial t+\nabla\cdot\mathbf{P}=0$.

7) The velocity field $\mathbf{u}(\mathbf{x},t)$ of an inviscid compressible gas obeys

$$\frac{\partial\rho}{\partial t}+\nabla\cdot(\rho\mathbf{u})=0 \quad\text{and}\quad \rho\left(\frac{\partial\mathbf{u}}{\partial t}+(\mathbf{u}\cdot\nabla)\mathbf{u}\right)=-\nabla p,$$

where $\rho(\mathbf{x},t)$ is the density and $p(\mathbf{x},t)$ is the pressure. Show that

$$\frac{\partial}{\partial t}\big(\tfrac{1}{2}\rho u_iu_i\big)+\frac{\partial}{\partial x_i}\big(\tfrac{1}{2}\rho u_ju_ju_i+pu_i\big)=p\,\nabla\cdot\mathbf{u} \quad\text{and}\quad \frac{\partial}{\partial t}(\rho u_i)+\frac{\partial t_{ij}}{\partial x_j}=0$$

for a suitable symmetric tensor t_{ij}, to be determined.

8) Using the transformation law for a second-rank tensor T_{ij}, show that the quantities

$$\alpha = T_{ii}, \qquad \beta = T_{ij}T_{ji}, \qquad \gamma = T_{ij}T_{jk}T_{ki}$$

are scalars, i.e., remain the same in all Cartesian coordinate systems. If T_{ij} is diagonal in some coordinate system, express the quantities above in terms of its eigenvalues. Hence deduce that the eigenvalues are the roots of the cubic equation

$$\lambda^3 - \alpha\lambda^2 + \tfrac{1}{2}(\alpha^2 - \beta)\lambda - \tfrac{1}{6}(\alpha^3 - 3\alpha\beta + 2\gamma) = 0.$$

9) If $\mathbf{u}(\mathbf{x})$ is a vector field, show that $\partial u_i/\partial x_j$ transforms as a second rank tensor field. If $\sigma_{ij}(\mathbf{x})$ is a tensor field, show that $\partial\sigma_{ij}/\partial x_j$ transforms as a vector field.

10) Note: Part (ii) may be clearer after some matrices lectures

(i) By first decomposing into symmetric and anti-symmetric parts, show that an arbitrary second rank tensor T_{ij} can be written in the form

$$T_{ij} = \alpha\delta_{ij} + \epsilon_{ijk}\omega_k + B_{ij}$$

where α is a *scalar*, $\boldsymbol{\omega}$ a *vector* and B_{ij} a symmetric traceless second rank *tensor*.

(ii) Let $\mathbf{u}(\mathbf{x})$ be a vector field and let $T_{ij} = \partial u_i/\partial x_j$. Show that $\alpha = \frac{1}{3}\nabla\cdot\mathbf{u}$ and $\boldsymbol{\omega} = -\frac{1}{2}\nabla\times\mathbf{u}$. Find B_{ij} in the case $\mathbf{u}(\mathbf{x}) = (xy^2, yz^2, zx^2)$: verify that $(0,0,1)$ is a principal axis at the point $\mathbf{x} = (2,3,0)$ and find the others.

11) Note: Part (ii) is trickier

(i) A tensor of rank 3 satisfies $T_{ijk} = T_{jik}$ and $T_{ijk} = -T_{ikj}$. Show that $T_{ijk} = 0$.

(ii) A tensor of rank 4 satisfies $T_{jikl} = -T_{ijkl} = T_{ijlk}$ and $T_{ijij} = 0$. Show that $T_{ijkl} = \epsilon_{ijp}\epsilon_{klq}S_{pq}$ where $S_{pq} = -T_{rqrp}$. [*Hint: Try to generalise a technique pertaining to antisymmetric second rank tensors.*]

12) The array T_{ijk} is such that $T_{ijk}S_{jk}$ is a vector for every symmetric second rank tensor S_{jk}. Show by counterexample that T_{ijk} need not be a tensor; but show also that $T_{ijk} + T_{ikj}$ must be.

13) In linear elasticity, the symmetric second rank stress tensor σ_{ij} (representing the forces on an elastic medium) depends on the symmetric second rank strain tensor e_{kl} (representing the corresponding deformation) according to $\sigma_{ij} = c_{ijkl}e_{kl}$. Assuming that c_{ijkl} is symmetric in k and l, explain why it must be a fourth rank tensor. For an isotropic medium, use the most general possible form for c_{ijkl} (which you may quote) to show that

$$\sigma_{ij} = \lambda\delta_{ij}e_{kk} + 2\mu e_{ij},$$

where λ and μ are scalars. Invert this equation to express e_{ij} in terms of σ_{ij}, assuming that $\mu \neq 0$ and $3\lambda \neq -2\mu$. Explain why the principal axes of σ_{ij} and e_{ij} coincide.

14) Let $(x_0, x_1, x_2, x_3) \equiv (t, x_1, x_2, x_3)$ be coordinates for Minkowski space $\mathbb{R}^{1+3}$. Show that the two Maxwell equations $\nabla\times\mathbf{E} = -\partial\mathbf{B}/\partial t$, $\nabla\cdot\mathbf{B} = 0$, are equivalent to

$$\frac{\partial F_{ab}}{\partial x_c} + \frac{\partial F_{bc}}{\partial x_a} + \frac{\partial F_{ca}}{\partial x_b} = 0 \quad \text{where} \quad (F_{ab}) = \begin{pmatrix} 0 & E_1 & E_2 & E_3 \\ -E_1 & 0 & -B_3 & B_2 \\ -E_2 & B_3 & 0 & -B_1 \\ -E_3 & -B_2 & B_1 & 0 \end{pmatrix}.$$

15) In the following, the indices i, j, k, ℓ take the values $1, 2, 3$ and the summation convention applies.

(a) Simplify the following expressions:

$$\delta_{ij}v_j\,, \quad \delta_{ij}\delta_{jk}\,, \quad \delta_{ij}\delta_{ji}\,, \quad \delta_{ij}v_iv_j\,, \quad \varepsilon_{ijk}\delta_{jk}\,, \quad \varepsilon_{ijk}v_jv_k\,, \quad \varepsilon_{ijk}\varepsilon_{ij\ell}\,, \quad \varepsilon_{ijk}\varepsilon_{ikj}\,.$$

(b) Given that $A_{ij} = \varepsilon_{ijk}a_k$ (for all i, j), show that $2a_k = \varepsilon_{kij}A_{ij}$ (for all k).

(c) Show that $\varepsilon_{ijk}S_{ij} = 0$ (for all k) if and only if $S_{ij} = S_{ji}$ (for all i, j).